



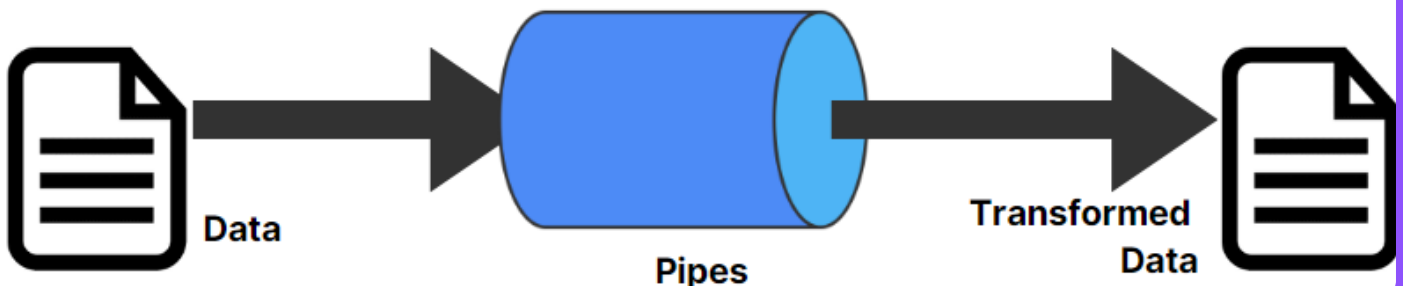
What are Pipes in Angular?

Pipes in Angular are a powerful feature that allows you to transform data directly within your templates. They take input data and format it into a desired output, making it easier to display information in a user-friendly way.

Key Features of Pipes

- Declarative Transformation

Pipes let you define transformations once and reuse them across multiple templates. Complete rewrite of AngularJS, released in 2016.





- **Built-in Pipes**

Angular provides several built-in pipes, such as

- **DatePipe:**

Formats date values.

- **UpperCasePipe & LowerCasePipe:**

Converts text to uppercase or lowercase.

- **CurrencyPipe:**

Formats numbers as currency.

- **DecimalPipe:**

Formats numbers with decimal points.

- **PercentPipe:**

Converts numbers into percentage format.

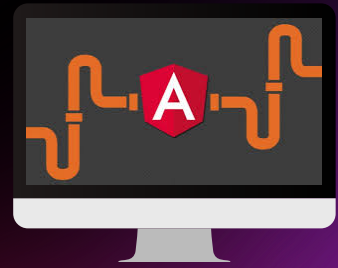
- **JsonPipe:**

Converts objects into JSON strings.

- **SlicePipe:**

Extracts a portion of an array or string.

```
1    <p>{{ today | date:'fullDate' }}</p>
2    <p>{{ 'hello world' | uppercase }}</p>
3    <p>{{ price | currency:'USD' }}</p>
```



Custom Pipe

- You can also create custom pipes using the **@Pipe** decorator to define your own transformation logic.

Step: 1 Generate the Pipe

- Use Angular CLI to create a pipe.

```
1 ng generate pipe custom
```

- This creates two files:
- custom.pipe.ts** and **custom.pipe.spec.ts**.

Step: 2 Implement the Pipe Logic

- Open **custom.pipe.ts** and define the **transformation logic**.

```
1 import { Pipe, PipeTransform } from '@angular/core';
2
3 @Pipe({
4   name: 'custom'
5 })
6 export class CustomPipe implements PipeTransform {
7   transform(value: string): string {
8     return `★ ${value} ★`; // Adds stars around the text
9   }
10 }
```



Step: 3 Use the Pipe in a Template

- Apply the **pipe** in an Angular component.

```
1 <p>{{ 'Hello Angular' | custom }}</p>
```

- This will display: ★ Hello Angular ★

Step: 4 Register the Pipe

- If using standalone components, import the pipe directly.

```
1 import { CustomPipe } from './custom.pipe';
2
3 @Component({
4   selector: 'app-example',
5   template: `<p>{{ 'Angular' | custom }}</p>`,
6   standalone: true,
7   imports: [CustomPipe]
8 })
9 export class ExampleComponent {}
```